MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATICS TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

COURSE: CLOUD COMPUTING

ASSIGNMENT: LAB 0

GROUP: DAZ

DAZ	1	Mwizerwa Timothy	2400724330	24/U/24330/PS
	2	Kinda Kelsey Naluwafu	2400705862	24/U/05862/PS
	3	Kato Adrian Mugerwa	2400725909	24/U/25909/PS
	4	Naluyima Pauline Oliver	2400708847	24/U/08847/PS
	5	Aroba Joseph	2400703646	24/U/03646/EVE

Lab 0 Report:
Setting up CloudLab Environment

Introduction

This lab aimed at helping us get started with the project infrastructure. The main goal was to ensure that all group members could access the required systems, set up accounts, and successfully run a basic CloudLab experiment. The lab also helped us understand teamwork, version control, and remote server access.

Setup and Collaboration Group

First, we confirmed our group members using the provided group spreadsheet. We also verified our group email (e.g., csc26-daz@cit.ac.ug) by sending a test email, and all members received it successfully. To improve collaboration, we created a private GitHub repository and added all group members as collaborators "https://github.com/TIMOTHY-ctrl/cloud computing".

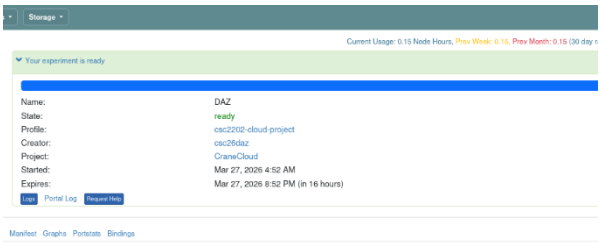
CloudLab Account Creation

We created a CloudLab account using the group email and group ID as instructed. The required details included username, full name, email, and affiliation (Makerere University Kampala). We also uploaded an SSH public key for secure access to CloudLab machines.

After submitting the request and verifying the email, we waited for account approval. Once approved, we logged into the CloudLab dashboard successfully.

Initial Configuration

After logging in, we checked and confirmed that our SSH keys were properly added. We also changed the default shell to **bash**, as recommended for running scripts. This step ensures compatibility with most project scripts.



Starting an Experiment

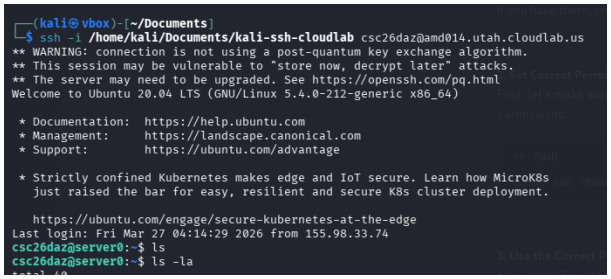
We started an experiment by selecting the project profile and choosing the Lab 0. We named the experiment using our group ID 'daz' and selected Utah cluster. After a few minutes, then experiment was successfully created.

The system automatically provided client and server machines. We accessed these machines using SSH commands provided on the dashboard once their status showed "Finished."

SSH Access and Authentication Setup

We used SSH to log into the CloudLab machines. To make access easier, we set up an SSH authentication agent on our local machines. This allowed us to connect to multiple servers without entering passwords repeatedly.

We used commands like ssh-agent, ssh-add, and ssh -A to enable secure and smooth access across all nodes.



Challenges Faced

- Delay in account approval (waiting time)
- Initial difficulty understanding SSH setup
- Minor issues with SSH key configuration

Conclusion: Overall, the lab was successful. We were able to set up our CloudLab environment, collaborate as a group, and run our first experiment. This lab provided a strong foundation for future project tasks, especially in cloud computing, teamwork, and server management.